Cyber Security

DNS Poisoning and counter measures

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https://techgirls.ece.vt.edu/slides/introduction_to_cybersecurity.html#39

DNS Security

Fundamental Problems of Network Security

- Internet was designed without security in mind
  - Initial design focused more on how to make it work, than on how to prevent abuses
  - Initial environment mostly consisted of research institutions---assumption on the benign behaviors of users
- Fundamental security problem of current network technology:
  - Has no way of telling whether the resource is located “correctly,” or the information is transferred “correctly”
  - Has no data authentication and confidentiality protection

Example Security Problems by Incorrect Resource Location

- DNS poisoning
- BGP routing vulnerabilities
- ARP poisoning
  - ARP (Address Resolution Protocol) is used to query for the MAC address associated with an IP address
  - Any device attached physically to a subnet can claim to be the “owner” of the IP
- IP Spoofing
  - Routers typically do not check source IP addresses
  - A packet can claim to be coming from any IP address
- Spam email

Fundamental Problems of TCP/IP

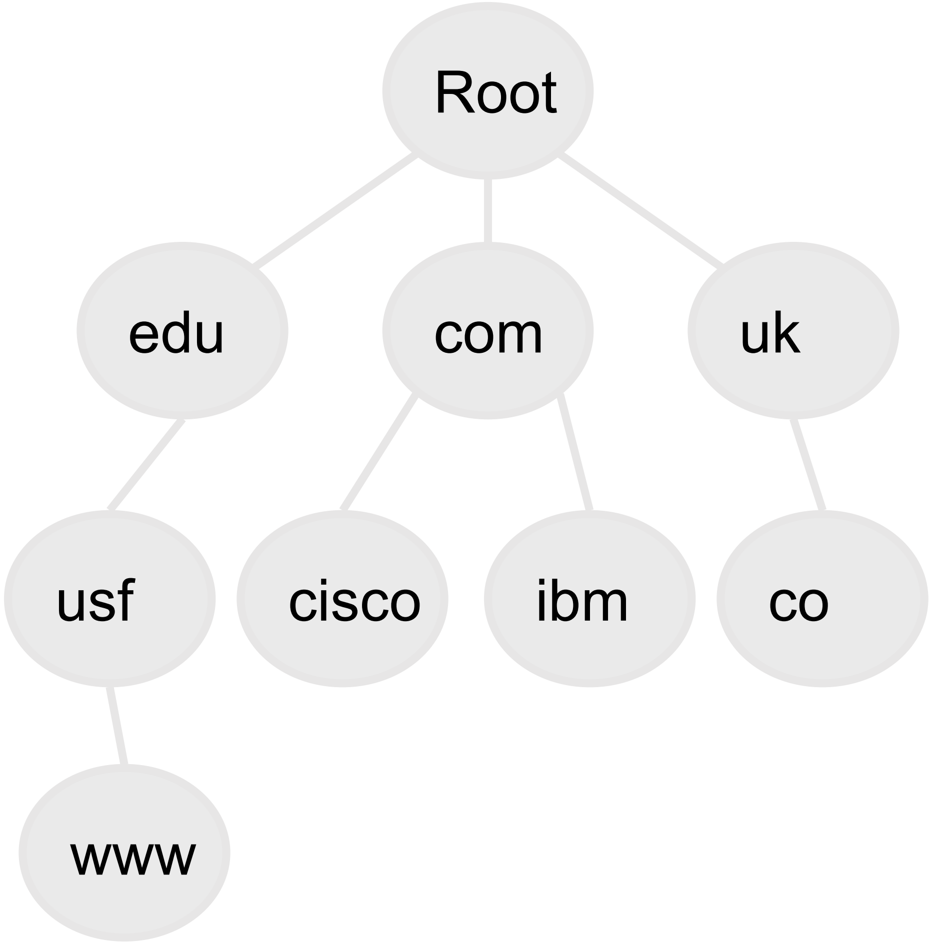
- No authentication for received messages

- No encryption for transmitted messages

- Applying cryptographic techniques can help
  - But must engineer very carefully

The Domain Name System

- Basic Internet Database
  - Maps names to IP addresses
  - Also stores IPv6 addresses, mail servers, service locators, Enum (phone numbers), etc.
- Data organized as tree structure.
  - Each ***zone*** is the authority for its local data.



Borrowed from slides of Prof. Dan Massey at Colorado State University

Domain Name Service

- Provides binding between URL and IP address
  - Both forward and reverse mapping
  - Divide URL space into zones; Each name server handles mapping in its zone

- DNS Resource Record (RR)
  - Can be viewed as tuples of the form

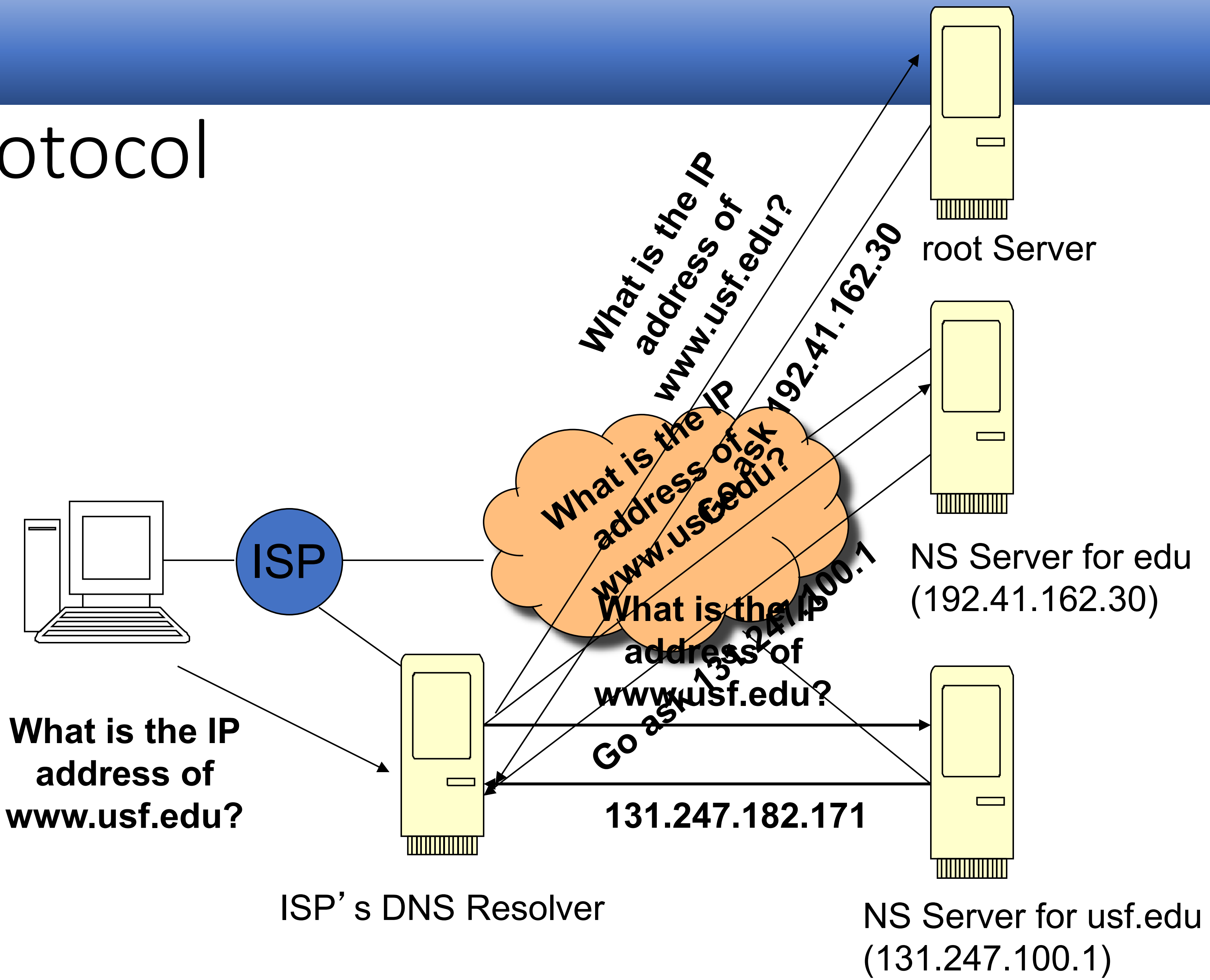
    <name, TTL, class, type, data>
  - types: A (IP address)

    MX (mail servers)

    NS (name servers)

    PTR (reverse look up)

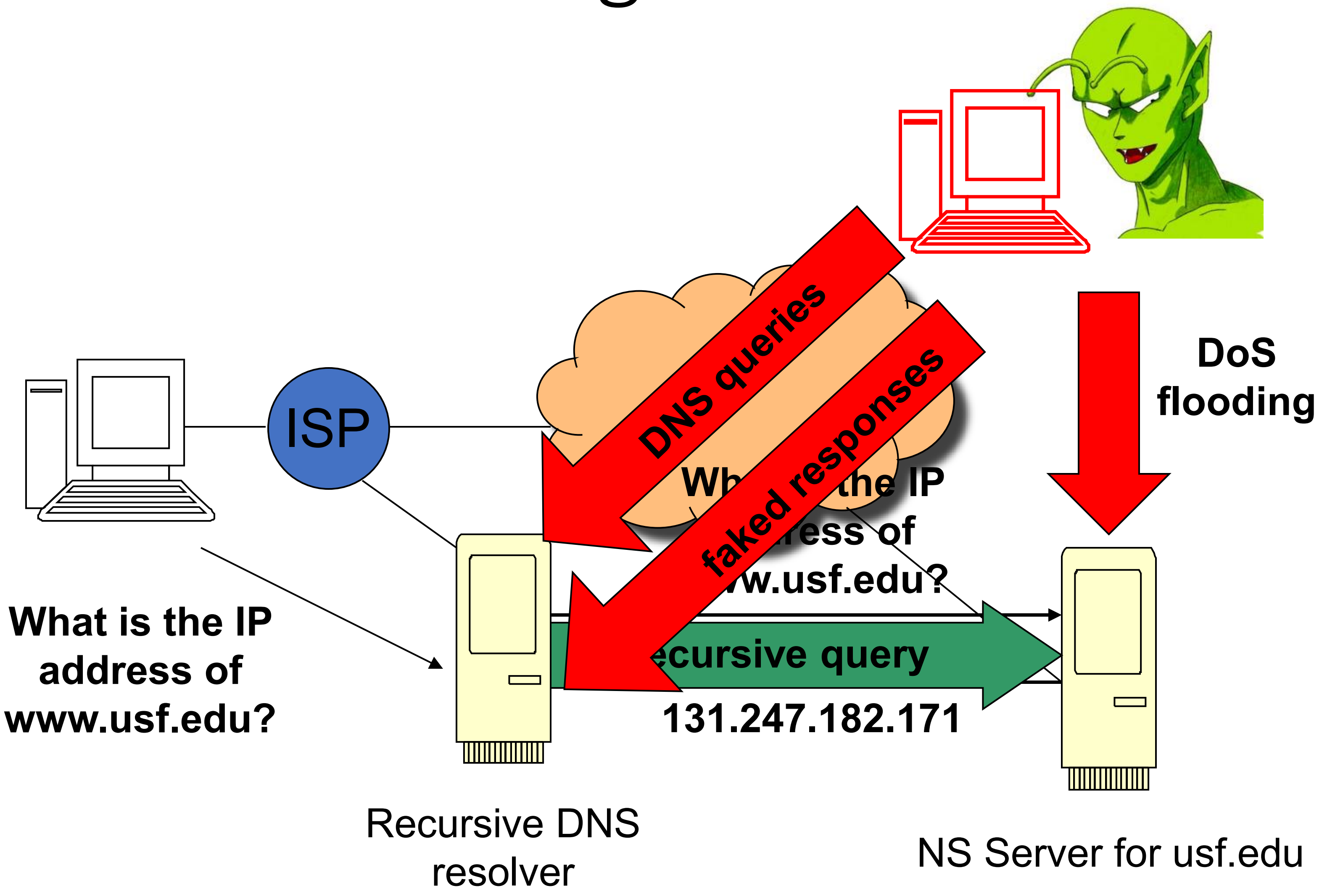
DNS Protocol



DNS Security Problems

- A DNS resolver has no way to determine if the response of a query does come from the legitimate server
- It will accept a response if
  - The port number matches the source port of the request
  - Has the correct Transaction ID (TXID).
- It will accept all RR's that are in the queried server's bailiwick
  - The bailiwick is the domain in which the server has authority according to the referral path

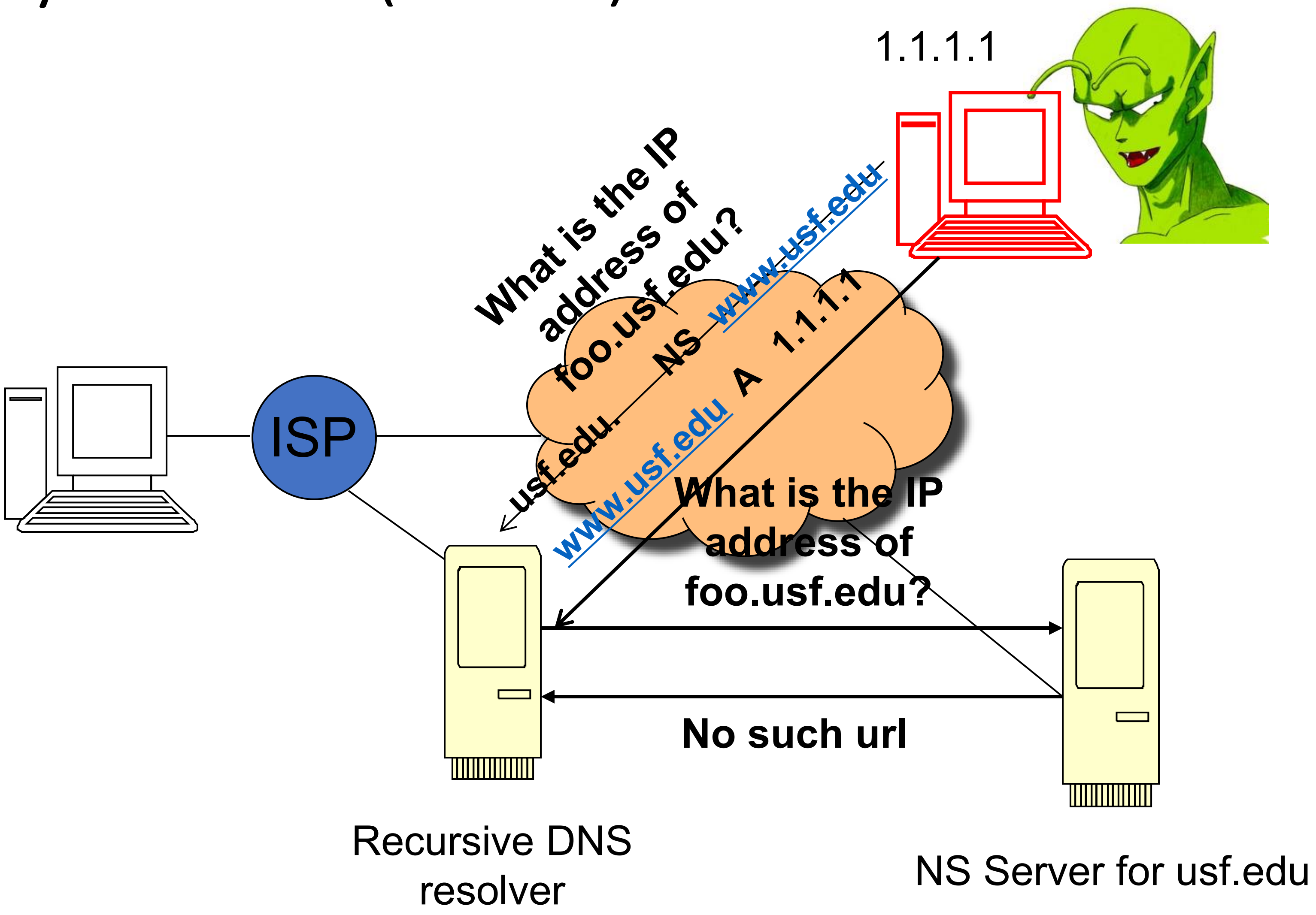
Classical DNS Poisoning



Conditions for classical DNS poisoning attacks

- Must guess right the correct source-port number
- Must guess right the correct TXID (16 bits)
- The fake response must arrive before the legitimate response
- If any of the above fails, the attack fails and there will be no chance to attack again until the TTL expires

Kaminsky Attack (2008)



Implication of Kaminsky Attack

- Dramatically reduces the complexity and increases the effectiveness of DNS cache poisoning
  - No longer needs to wait for TTL to expire
  - The attacker can control when and what queries are issued
  - A complete domain may be hijacked
    - Even TLD's are vulnerable
  - Only needs 10 secs to succeed

How to help prevent DNS poisoning

- Individuals can help prevent DNS poisoning by using a VPN, practicing online safety, changing DNS settings, and periodically clearing the DNS cache from devices — and scanning for malware in case they unwittingly fall victim to an attack.
- Website owners can help prevent DNS poisoning attacks by enabling DNS security extensions, configuring trusted DNS servers, and training employees.

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https://www.avast.com/c-what-is-dns-poisoning

Thank You